

EXPERIMENT 9: Rule-Based Part-of-Speech Tagging Using Regular Expressions

Aim: To implement a rule-based Part-of-Speech (POS) tagging system using regular expressions in Python.

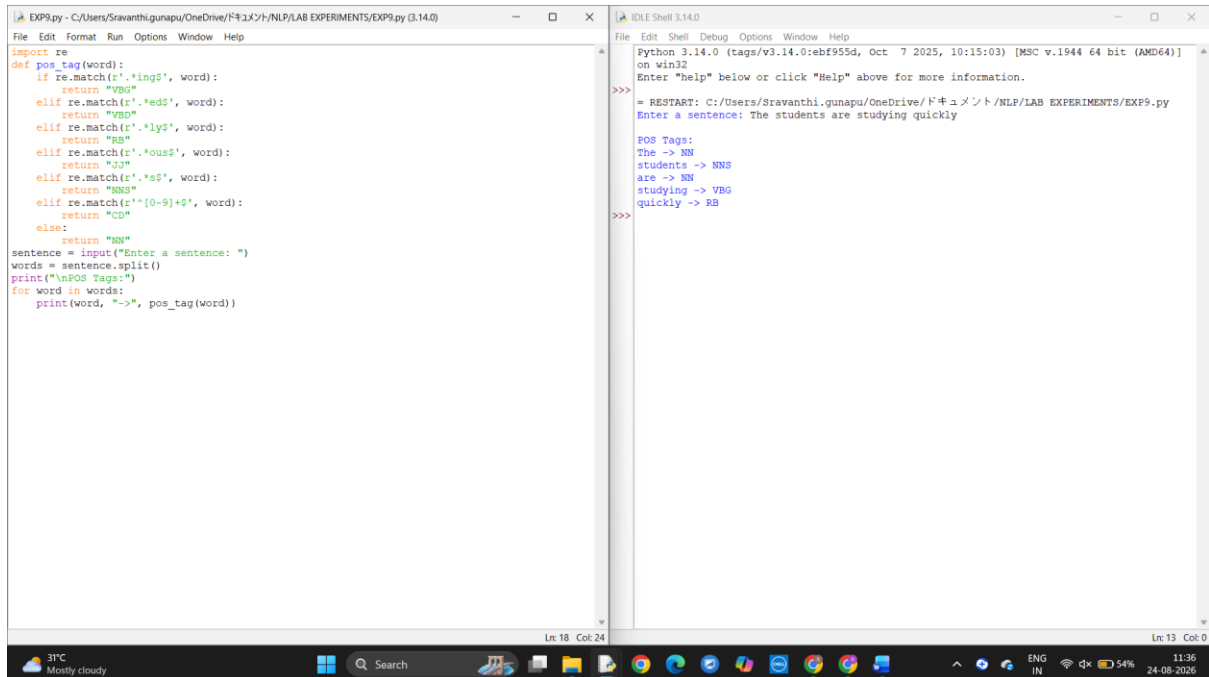
Program:

```
import re

def pos_tag(word):
    if re.match(r'.*ing$', word):
        return "VBG"
    elif re.match(r'.*ed$', word):
        return "VBD"
    elif re.match(r'.*ly$', word):
        return "RB"
    elif re.match(r'.*ous$', word):
        return "JJ"
    elif re.match(r'.*s$', word):
        return "NNS"
    elif re.match(r'^[0-9]+\$', word):
        return "CD"
    else:
        return "NN"

sentence = input("Enter a sentence: ")
words = sentence.split()
print("\nPOS Tags:")
for word in words:
    print(word, "->", pos_tag(word))
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'EXP9.py', contains a script for Part-of-Speech (POS) tagging. The script uses regular expressions to identify various POS tags based on word patterns. The right window, titled 'IDLE Shell 3.14.0', shows the execution of the script. It prompts the user to 'Enter a sentence:', and the input 'The students are studying quickly' is provided. The output shows the POS tags assigned to each word in the sentence: 'The' is tagged as NN, 'students' as NNS, 'are' as VB, 'studying' as VBG, and 'quickly' as RB.

```
import re
def pos_tag(word):
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        return "VBG"
    elif re.match(r'ed$', word):
        return "VBD"
    elif re.match(r'ly$', word):
        return "RB"
    elif re.match(r'ous$', word):
        return "JJ"
    elif re.match(r'is$', word):
        return "VBZ"
    elif re.match(r'^[0-9]+$', word):
        return "CD"
    else:
        return "NN"
sentence = input("Enter a sentence: ")
words = sentence.split()
print("\nPOS Tags:")
for word in words:
    print(word, "->", pos_tag(word))
```

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP9.py
Enter a sentence: The students are studying quickly

POS Tags:
The -> NN
students -> NNS
are -> VB
studying -> VBG
quickly -> RB

>>>
```